Ind: City?

Machine → Keep Consuming at power rating

if PF low (Volt fixed) → 1 MW → 440 V / 880 V

fixed → P = V_{rms} I_{rms} PF (fixed)

↳ Curr (RMS) → ↑ to match neg pwr

I^2R

Tut 9 Prob 6

4 kW = V_{rms} I_{rms} $\cos\theta$ → 220

$\cos\theta = 0.8$

$\theta = \tan^{-1}\left(\frac{X_L}{R_L}\right)$

220 V AC

R_L, θ, X_L, $|Z_L|$

$Z_L = R_L + jX_L$, $X_L > 0$

4 kW, PF 0.8 lagging

How? PF = 0.9

Mort

$I_{rms} = \frac{V_{rms}}{|Z_L|}$

$|Z_L|$ ↓, θ ↓, X_L ↓

Household — Unit — charged based on 1 kWh

Industrial — Pay ₹ per 1 kWh — same for penalized if PF < 0.8

500 W Ind heater PF ≈ 0.9

220 . I_{rms} 0.55 = 5 kW

$I_{rms} \simeq 24A$

Ex: 11-10

$\left(\frac{jw}{2}+1\right)\left(\frac{jw}{100}+1\right)$

$\rho e^{j\theta}$ $\rho' e^{j\theta'}$

$\tan^{-1}\frac{w}{2} + \tan^{-1}\frac{w}{100} - \tan^{-1}\frac{w}{30} - \tan^{-1}\frac{w}{300}$

$$H(s) = \frac{\left(\frac{s}{2}+1\right)\left(\frac{s}{100}+1\right)}{\left(\frac{s}{30}+1\right)\left(\frac{s}{300}+1\right)}$$

H_{dB}

$20\log|H(jw)|$

+ 2nd s/de.

$\pi/2$

$\pi/4$

+20dB

2 30 100

- 20dB

$\log w \rightarrow$

$-\pi/4$

$-\pi/2$

Phase =

$\frac{Z_f(s)}{Z_i(s)}$

$Z_f(s)$

$i=0$

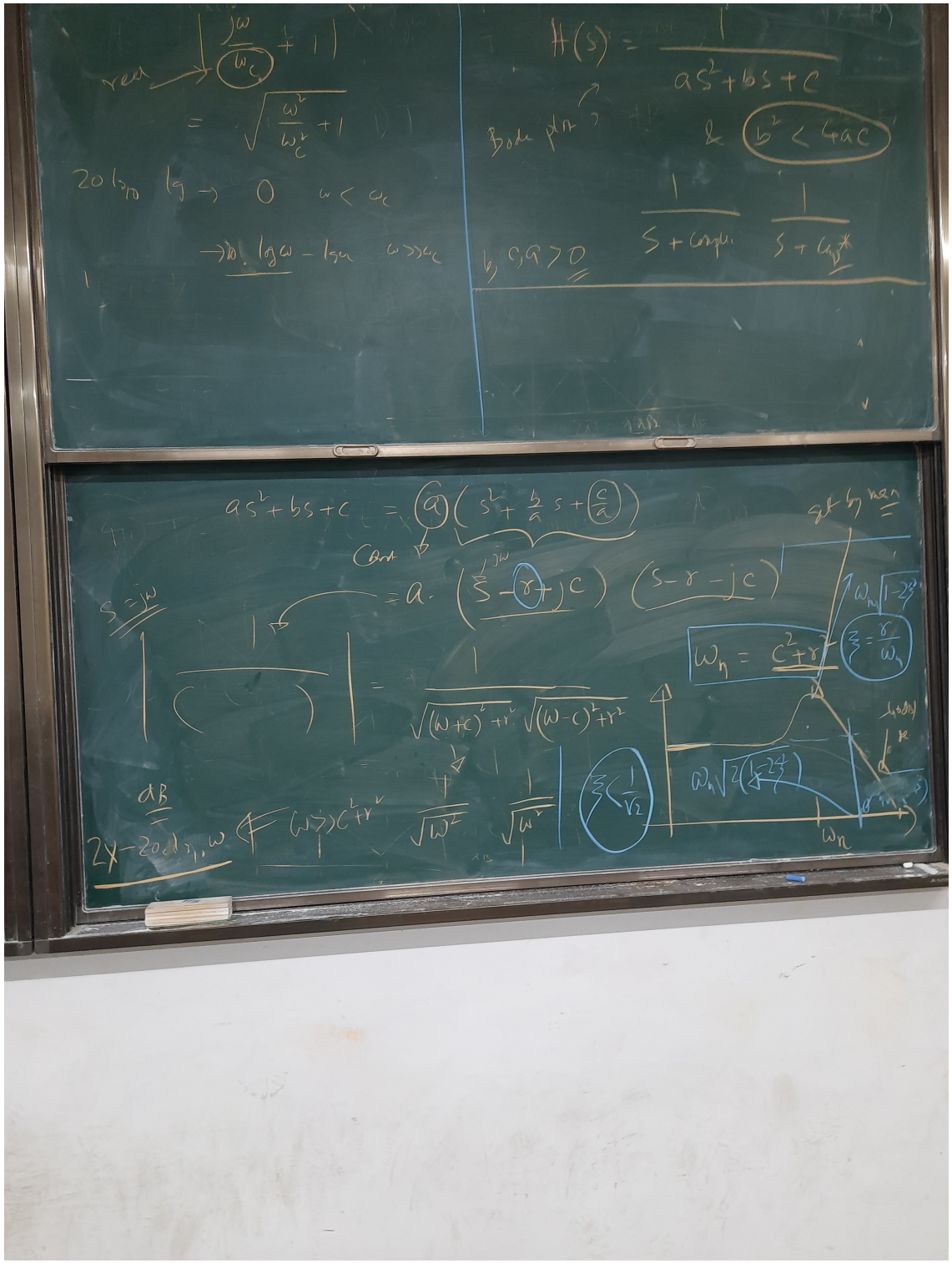
$Z_i(s)$

G_M

V_{in}

V_{out}

$H(s) = \frac{Z_f(s)}{Z_i(s)}$


$\left|\frac{j\omega}{\omega_c}+1\right|$
real
$=\sqrt{\frac{\omega^2}{\omega_c^2}+1}$
$20\log_{10}$ lg → 0 $\omega<\omega_c$
→ 20 log ω − log ω_c $\omega\gg\omega_c$
$H(s)=\frac{1}{as^2+bs+c}$
Bode plot
& $b^2<4ac$
$\frac{1}{s+\text{complex}}$ $\frac{1}{s+\text{complex}^*}$
by $a,a>0$
$as^2+bs+c=a\left(s^2+\frac{b}{a}s+\frac{c}{a}\right)$
Const
$=a\cdot(s-\gamma+jc)(s-\gamma-jc)$
set by man
$s=j\omega$
$\omega_n=\sqrt{c^2+\gamma^2}$
$\omega_n\sqrt{1-\xi^2}$
$\xi=\frac{\gamma}{\omega_n}$
$=\frac{1}{\sqrt{(\omega+c)^2+\gamma^2}\sqrt{(\omega-c)^2+\gamma^2}}$
dB
$2\times-20\log_{10}\omega$ if $\omega\gg c^2+\gamma^2$
$\sqrt{\omega^2}$ $\sqrt{\omega^4}$
$\xi<\left(\frac{1}{\sqrt{2}}\right)$
$\omega_n\sqrt{2(1-2\xi^2)}$
ω_n